Enrollment No._____

GANPAT UNIVERSITY

B.Sc.IT(DS) Sem-V Regular/Remedial Nov-Dec 2024

U75A1PYT Python Programming

Time: 3 Hours | Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BTL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy**
1	What are the built-in data types in Python? Give examples of each.	06	CO1	BTL3
2	What is a set in Python? How is it different from other data types like lists and tuples?	06	CO1	BTL4
3	What is the difference between a function and a method in Python? Provide examples.	06	CO2	BTL4
4	What is list slicing in Python? Write an example that slices a list to get a subset of its elements.	06	CO3	BTL6
5	What is the with statement in Python? Explain its use in file handling with an example.	06	CO4	BTL3
6	What are the different types of files in Python? Explain the difference between text files and binary files.	06	CO4	BTL4
7	What is NumPy, and how does it support scientific computing? Write an example of creating a NumPy array.	06	CO5	BTL5
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy**
1	What are the different types of loops in Python? Write a simple example using a for loop and a while loop.	06	CO1	BTL4
2	What are positional and keyword arguments in Python? Write a function that demonstrates both.	06	CO2	BTL3
3	How are function decorators used in Python? Provide a simple example of a function decorator.	06	CO2	BTL5
Q.3	**Answer the following: (Any Two)**	**12**		
1	What is list traversal in Python? Write a program that traverses and prints each element of a list.	06	CO3	BTL4
2	How do you create a dictionary in Python? Write an example of a dictionary with three key-value pairs.	06	CO3	BTL6
3	What are the seek() and tell() methods in Python file handling? Write a program to demonstrate their usage.	06	CO4	BTL5
Q.4	**Answer the following: (Any One)**	**06**		
1	What is Matplotlib, and what is it used for in Python? Write a program to plot a simple line graph.	06	CO5	BTL6
2	What is Pandas, and why is it used for data analysis? Write an example of creating a DataFrame from a dictionary.	06	CO5	BTL5

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

B.Sc. IT (Data Science) – V (CBCS)

Remedial \ Regular – Nov. - Dec. 2024

U75A2IRP – Introduction to R Programming

Time: 3 Hours

Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

	Section – I			
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy Pattern**
1	Define R Programming. Discuss the advantages of R Programming.	06	CO1	BTL1
2	Explain vectors in R programming. How can you create, access and sort vector elements?	06	CO2	BTL6
3	How can you create Data frame in R? differentiate between Data frame and Matrix.	06	CO2	BTL4
4	Explain ways to create single & multidimensional array in R?	06	CO2	BTL6
5	Explain different statistical analysis functions of R.	06	CO3	BTL2
6	How can you read and analyze data from excel in R?	06	CO4	BTL3
7	Prepare pie chart in R using ggplot2. Customize the chart.	06	CO4	BTL6
	Section – II			
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy Pattern**
1	Explain various operators of R programming with example.	06	CO1	BTL1
2	How can you create function in R? Give proper example.	06	CO1	BTL6
3	What is package in R? how can you install and use package in R programming?	06	CO2	BTL3
Q.3	**Answer the following: (Any Two)**	**12**		
1	Explain the rep() and seq() functions with example.	06	CO2	BTL2
2	What is matrix in R? explain various functions like rbind(), cbind(), rowSums(), colSums() with example.	06	CO3	BTL6
3	Explain dplyr package of R with suitable example.	06	CO3	BTL5
Q.4	**Answer the following: (Any One)**	**06**		
1	How can you create and format Bar graphs in R?	06	CO4	BTL6
2	Differentiate between Bar graphs and Histograms. How can you create Histograms in R?	06	CO4	BTL4

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

B.Sc. IT (Data Science) Sem-V Reg\Rem Nov-Dec 2024

U75A4LAP:Linear Algebra and Probability Distribution

Total: 60 Marks

Time: 3 Hours

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BTL Pattern Level.
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3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I

		Marks	CO	Bloom's Taxonomy
Q-1	**Answer the following. (Any 5)**	**30**		
1	Solve following system by Gaussian elimination method, 1) $x + y + 2z = 9$ 2) $2x + 4y - 3z = 1$ 3) $3x + 6y - 5z = 0$	06	CO1	BTL3
2	If $A = \begin{bmatrix} 1 & 2 & 1 \\ a & 0 & 4 \\ 1 & 1 & 1 \end{bmatrix}$ and $Adj(Adj\ A) = A$ then find a.	06	CO1	BTL2
3	Solve the following system of equations by matrix inversion method. (1) $x + y + z = 8$, (2) $x - y + 2z = 6$ and (3)$3x + 5y - 7z = 14$	06	CO1	BTL3
4	Define Vector space V over field F. Then show that $W = \{(x, y) : x = 3y\}$ is a subspace of $\mathbb{R}^2$.	06	CO2	BTL2
5	State and prove Pythagorean theorem.	06	CO3	BTL1
6	Find eigen values of $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$.	06	CO4	BTL5
7	Let x be continuous random variable with probability density function given by $f(x) = \begin{cases} ax\,; 0 \le x \le 1 \\ a; 1 \le x \le 2 \\ -ax + 3a\,; 2 \le x \le 3 \\ 0; 3 < x \end{cases}$ (1) Determine constant a. (2) Calculate $P(x \le 1.5)$.	06	CO5	BTL5

Section – II

		Marks	CO	Bloom's Taxonomy
Q-2	**Answer the following (Any Two)**	**12**		
1	Investigate for what values of λ and μ equation $x + 2y + z = 8$, $2x + 2y + 2z = 13$, $3x + 4y + \lambda z = \mu$ have (1) no solution ,(2) unique solution, (3)infinite solution.	06	CO1	BTL4
2	Find rank and nullity of the matrix, $A = \begin{bmatrix} 2 & 0 & -1 \\ 4 & 0 & -2 \\ 0 & 0 & 0 \end{bmatrix}$.	06	CO2	BTL5
3	Check that (0,4,5) is a linear combination of $v_1 = (0,-2,2)$and $v_2 = (1,3,-1)$ or not.	06	CO2	BTL4
Q.3	**Answer the following: (Any Two)**	**12**		
1	Let $V = \mathbb{R}^n(\mathbb{R})$ and $\alpha = (x_1, x_2, x_3, \dots\dots, x_n)$ and $\beta = (y_1, y_2, y_3, \dots\dots, y_n)$ define $< \alpha, \beta >= \sum_{i=1}^{n} x_i y_i$. Show that $(V, <,>)$ is an inner product space. If $\alpha = (1,2,3)$ and $\beta = (-1,4,9)$ then find $< \alpha, \beta >$.	06	CO3	BTL5
2	Let $\mathbb{R}^3$ have the Euclidean inner product space. Use the Gram-Schmidth process to transform the basis of vectors $u_1 = (1,0,0)$,$u_2 = (3,7,-2)$, $u_3 = (0,4,1)$ into an orthonormal basis.	06	CO3	BTL6
3	Determine the nature (value class), index and signature of the following quadratic forms: A. $x_1^2 + 5x_2^2 + x_3^2 + 2x_2x_3 + 6x_3x_1 + 2x_1x_2$ B. $x_1^2 + 4x_2^2 + x_3^2 - 4x_2x_3 + 2x_3x_1 - 4x_1x_2$	06	CO3	BTL5
Q.4	**Answer the following: (Any One)**	**06**		
1	From a city population, the probability of selecting (1) a male or a smoker is $\frac{7}{10}$. (2) a male if a smoker is already selected is $\frac{2}{3}$. Then find probability of selecting (a) non-smoker, (b) a male, (c) a smoker if a male is first selected.	06	CO4	BTL5
2	Two balls are selected at random from a box containing three red, two green and four white. If X and Y are the number of red balls and green balls respectively included among the two balls drawn from the box find marginal probability distribution of X and Y .	06	CO4	BTL6

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

B.Sc. IT(DS) Sem-V Reg\Rem Nov-Dec 2024

U75B6ELS: Professional Ethics and Life Skills

Time: 3 Hours Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BTL Pattern Level.
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6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy**
1	Define ethics and explain its role in personal development.	06	CO1	BTL1
2	Compare and contrast the concepts of caring and sharing.	06	CO1	BTL2
3	Explain the concept of moral autonomy and its relevance.	06	CO2	BTL4
4	Define safety and risk in a professional environment.	06	CO3	BTL1
5	Develop a safety assessment plan that considers both employee and professional rights.	06	CO3	BTL6
6	Identify different types of values and their impact on personal growth.	06	CO4	BTL1
7	Assess the importance of security and survival for community welfare	06	CO4	BTL3
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy**
1	Describe the importance of integrity in a professional setting.	06	CO1	BTL3
2	List the types of moral inquiries in professional ethics.	06	CO2	BTL2
3	Discuss the role of ethical theories in making right actions.	06	CO2	BTL5
Q.3	Answer the following: (Any Two)	**12**		
1	Explain the importance of confidentiality in professional settings.	06	CO3	BTL4
2	valuate the ethical considerations around intellectual property rights.	06	CO4	BTL5
3	Analyze the role of humility and sympathy.	06	CO4	BTL4
Q.4	**Answer the following: (Any One)**	**06**		
1	Define society and list its basic units.	06	CO4	BTL1
2	Compare ancient and modern societies in terms of social responsibility.	06	CO4	BTL3

**** End of the Paper ****